

Human Computer Interaction

Johan F. HOORN and Jeff TANG

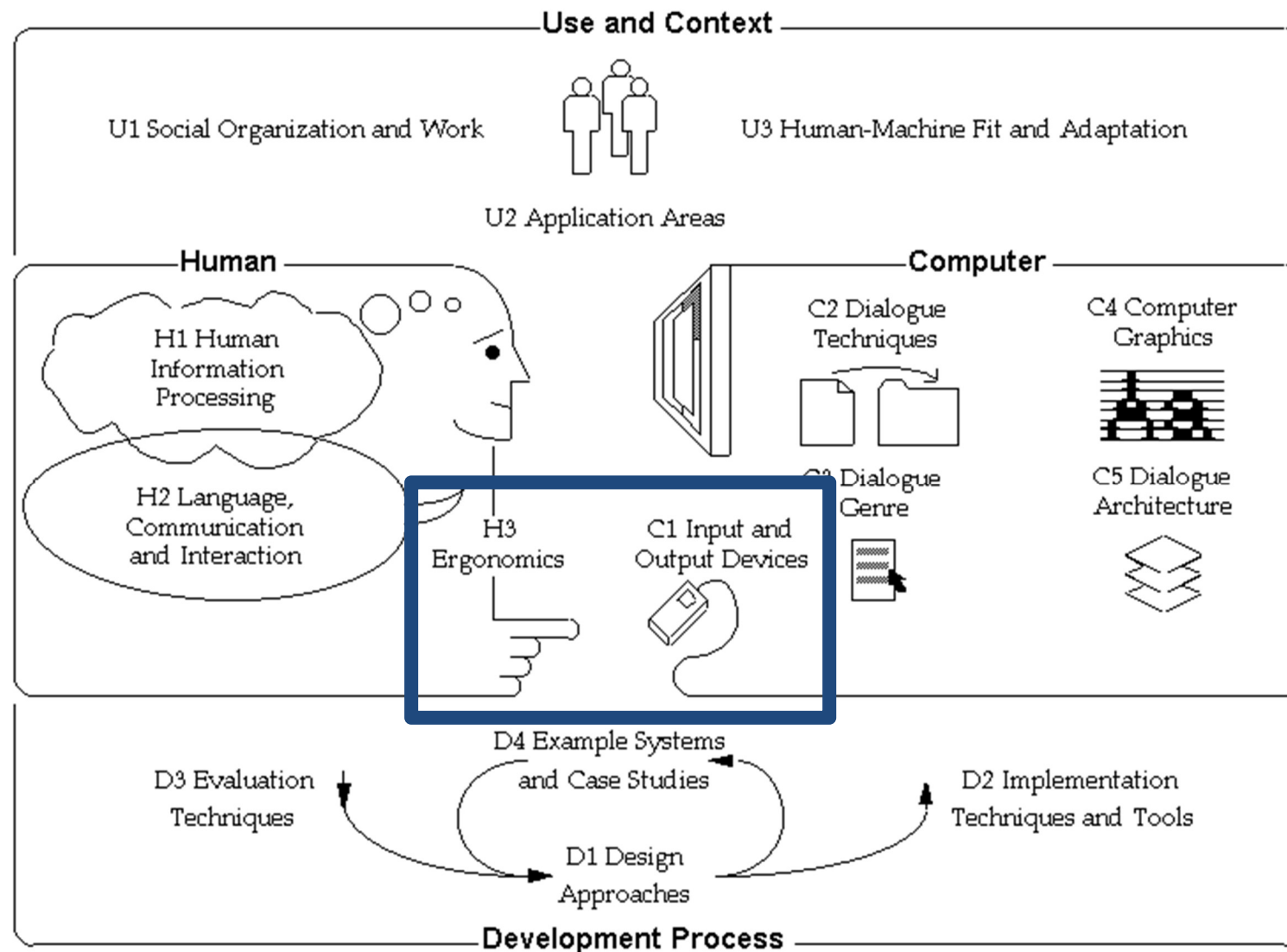
Lecture 05, Part 1: Ergonomics and I/O devices



C-3PO ("3PO" aka Threepio)

Lukas, G., & Hales, J. (2002). *Star Wars Episode II: Attack of the Clones*. Los Angeles, CA: 20th Century Fox. [Part 7/10] - 1080p HD, 7:00 min. Retrieved May 21, 2012 from http://www.youtube.com/watch?v=-o5Y_AszZGg

Overview



Contents

- ❑ Ergonomics
- ❑ Input Devices
- ❑ Output Devices
- ❑ Haptics
- ❑ Sound



Slides in this lecture are partially based on those from Ms. Tiffany Tang, Dr. Vincent Ng, Dr. Grace Ngai, Dr. Saul Greenburg, David Ellsworth, Jelle vd Berg, Franz J. Kurfess, and Dr. A. K. Sengupta

Ergonomics, the definition

Ergonomics (or human factors) is the scientific discipline concerned with the understanding of the interactions among human and other elements of a system, and the profession that applies theory, principles, data and methods to design in order to optimize human well-being and overall system performance.

International Ergonomics Association (IEA) Executive Council

- Design for humans in interaction with software and hardware
- So to optimize human well-being and overall system performance



Ergonomics



Human Factors

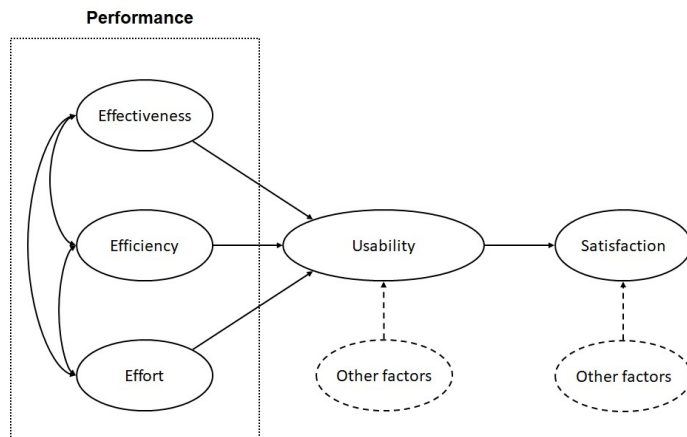
Ergonomics originally comes from the Greek words “ergon” (work or labor) and “nomos” (natural laws).

Ergonomics, refinement of the definition

❑ Overall system performance

Please remember from previous classes:

- The 'system' is both the human and the technical system in interaction (not the technology alone)
- Performance is determined by Effectiveness, Efficiency, and Effort (of both humans and machines)
- Efficiency is a trade-off between speed and accuracy (cf. Fitts' Law)
- 'Optimal' according to human judgment (users and other stakeholders)



Hoorn, J. F. (2014). Stakeholder logistics of an interactive system. *Journal of Systems and Software*, 95, 52-69. doi:10.1016/j.jss.2013.11.1095

Ergonomics, origins

❑ Governmental

- Military (e.g., World War II, the cockpits of fighter jets)

Remember? Prof. Paul Morris Fitts was Lieutenant-colonel of the US Air Force

Fitts' law states that the amount of time required for a person to move a pointer to a target area (e.g., finger to button) is a function of the distance to the target divided by the size of the target. Thus, the longer the distance and the smaller the target's size, the longer it takes

Thus, Fitts' Law relates to the Efficiency of system performance

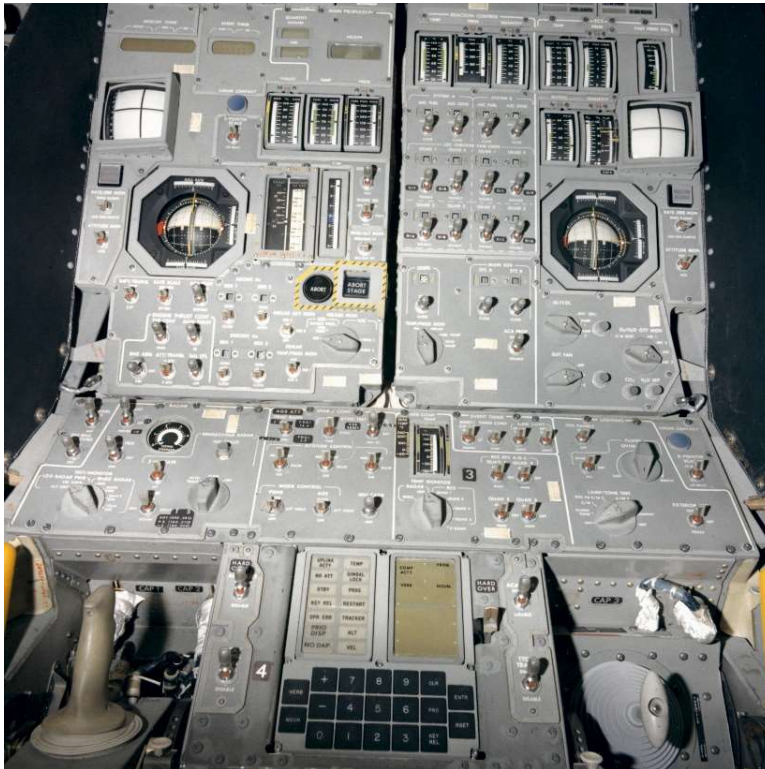


Cockpit controls of a Spitfire fighter

Ergonomics, origins

❑ Governmental

- Space exploration (e.g., the race for space during the Cold War)



Source: NASA



Source: SpaceX

How would Fitts' Law play out in space, without gravitation?

Ergonomics, origins

❑ Industry and workplace

After WW II, the focus is also on worker safety and productivity

- Muscle force required to perform manual tasks
- Compressive low-back disk-force when lifting
- Cardiovascular response when performing heavy labor
- Perceived maximum load that can be carried, pushed or pulled

● Automotive

- Car assembly lines (the workers)
- But also driving a car (the users)

Gkikas, N. (Ed.). (2012). *Automotive ergonomics: driver-vehicle interaction*. CRC Press.



1960s Jaguar: more tactile



1980s Lancia Orca: more visual



Today's Tesla: speech

Ergonomics, origins

❑ Industry and workplace

After WW II, the focus is also on worker safety and productivity

- To relieve back pain (when sitting still all day)
- To avoid slumping forward
- To burn calories (overweight causes heart disease)
- To relieve the eyes (when staring into a screen all day)

● Consumer products



Ergonomics, origins

❑ Industry and workplace

After WW II, the focus is also on worker safety and productivity

- No more green letters on black background
- No more yellow letters on blue background
- But black letters on white background (cf. printed matter)

• Computers



IBM with monochrome monitor

This adjustable desktop mount is to find the right angle for your head and eyes

Ergonomics, origins

- ❑ Disasters leading to lawsuits and public questions

Courts start calling for experts to explain human behavior, responses, defective design, and effectiveness of workplace warnings and instructions

- Three Mile Island (see Lecture 1)
- Chernobyl
- AF447 aviation disaster (next slide)





Ergonomics: physical and cognitive

❑ Physical

- Who are the users?
- How does technology fit (e.g., user dimensions, anatomy, strength, abilities)?
- How safe is the technology (health, comfort, performance)?
- How do users interact with technology?

❑ Cognitive

- How do users expect the technology to work?
- How is information displayed?
- How well are stereotypical expectations met?
- How complex is the interface?
- How much training is required?
- What user knowledge assumptions are met?
- How does information facilitate learning and memory?



“It’s an ergonomic keyboard. If your typing slows down, it automatically injects coffee into your fingertips.”

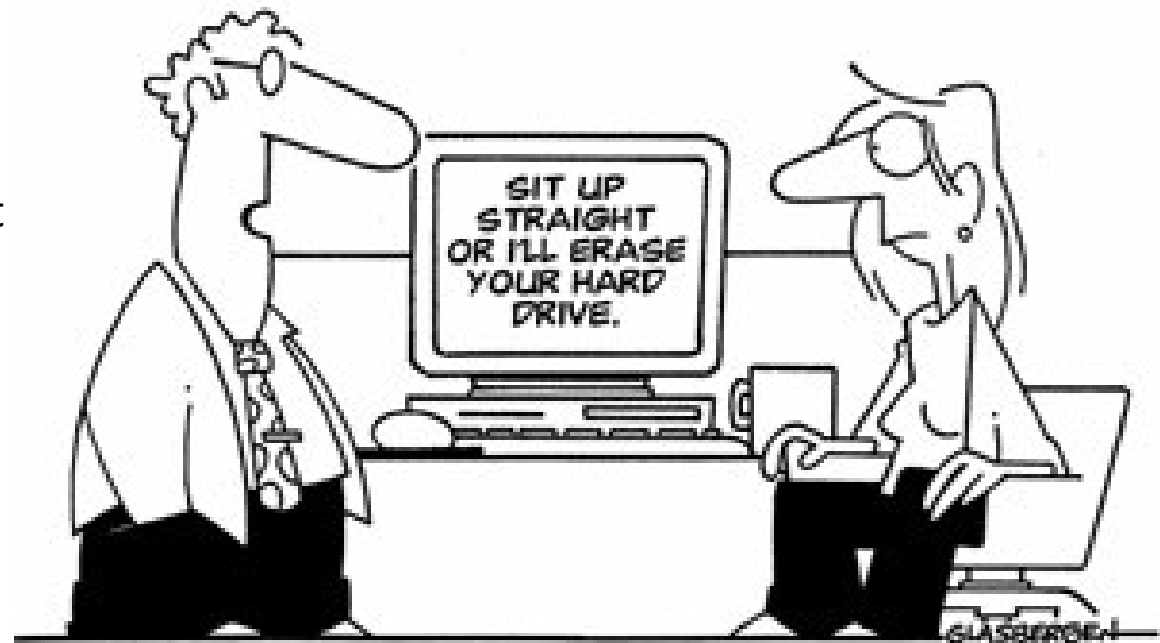
Physical Ergonomics

Physical Ergonomics is concerned with human anatomy, anthropometry (the measurement of human aspects), physiology and bio mechanical characteristics as they relate to physical activity in work or daily life

Certain jobs or work conditions cause a higher rate of worker complaints:

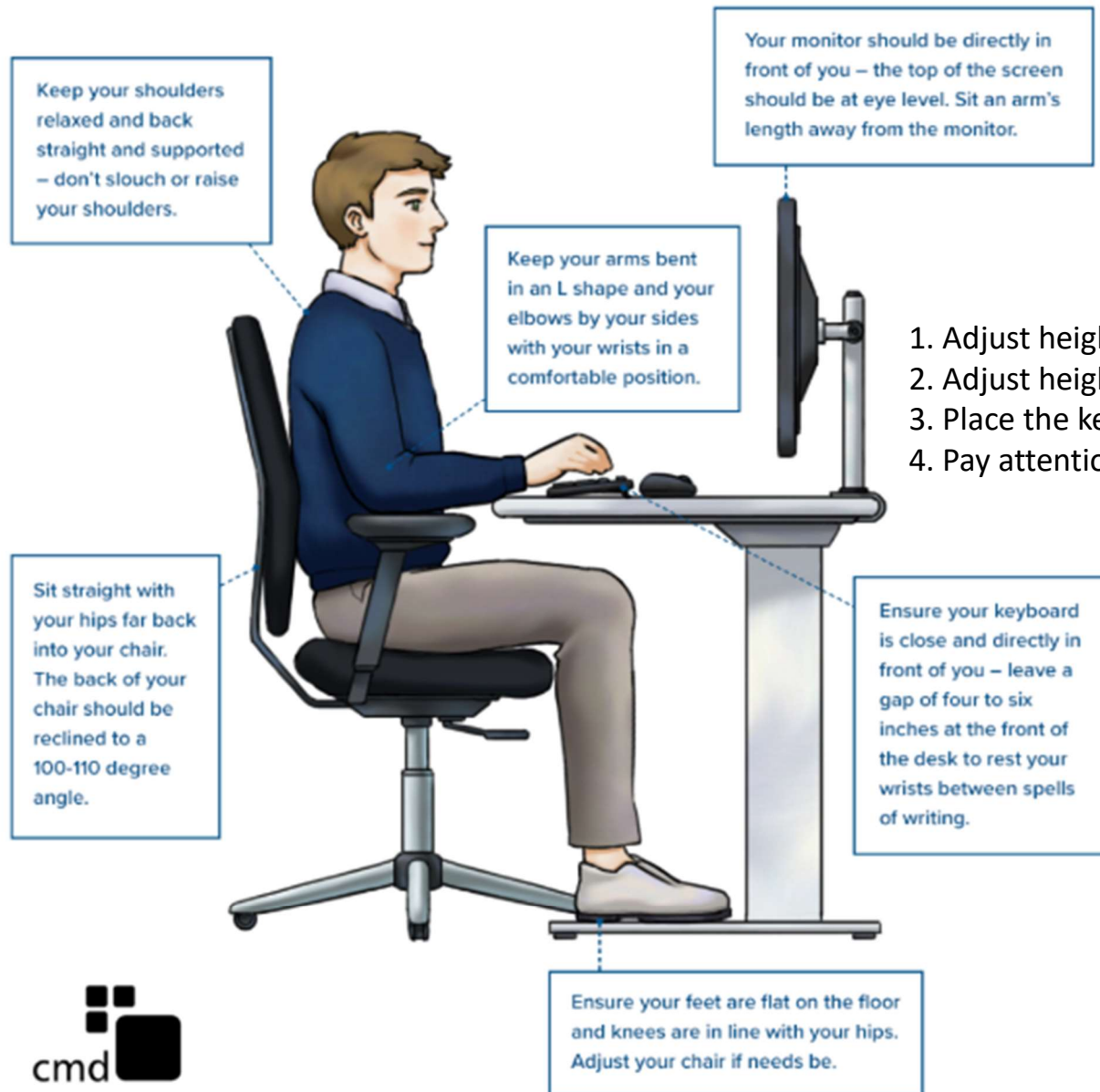
- Undue strain
- Localized fatigue
- Discomfort
- Pain that does not go away after overnight rest

© 2008 Randy Glasbergen. www.glasbergen.com



"It's called Ergonomics."

Physical Ergonomics

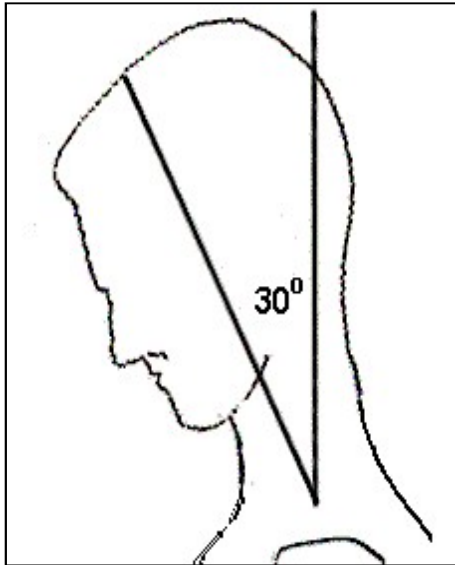


1. Adjust height of the chair & desk (if possible)
2. Adjust height of the monitor
3. Place the keyboard and mouse
4. Pay attention to lighting

Physical Ergonomics

Working with the neck or back bent forward more than 30° for more than 2 hours per day causes injuries

Source of graphics: OSHA



- Your head weighs 10 to 12 pounds, so it can be quite a strain to work with your neck bent
- Your upper body is 60 percent of your total body weight, so like your neck holding up your head, it's also a strain for your back to hold you in a bent over position for more than two hours per day
- Like neck posture, back posture is only covered if the worker is not able to vary his or her posture

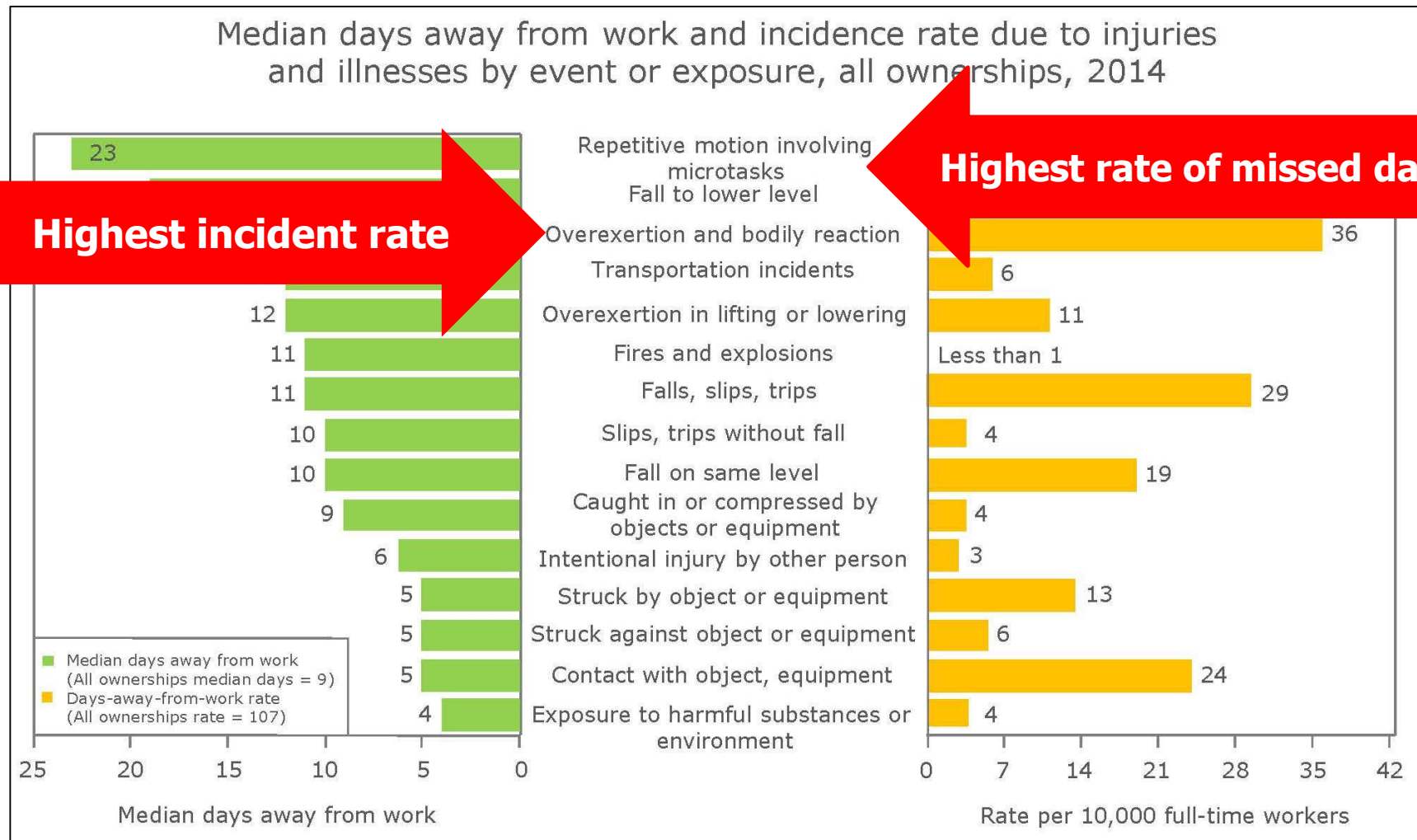
Physical Ergonomics

It's like holding a bowling ball for two hours



Source: stock photo from 123RF.com

Physical Ergonomics



Source: Bureau of Labor Statistics

Physical Ergonomics

❑ Musculo-skeletal disorders: Carpal Tunnel Syndrome

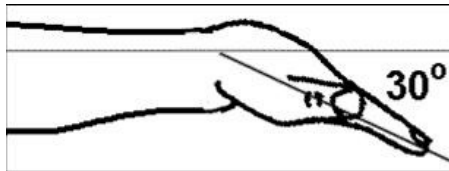
- Compression of the median nerve at the wrist
- Carpal tunnel made up of nine flexor tendons and one peripheral nerve
- Numbness and tingling on the thumb side of the hand



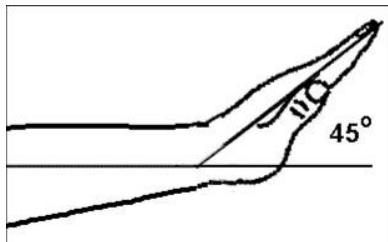
David Ellsworth

Physical Ergonomics

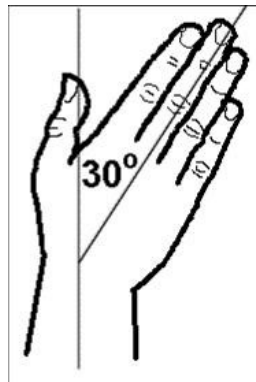
- ❑ Awkward postures, including wrists bent in flexion 30° or more, or in extension 45° or more, or in ulnar deviation 30° or more



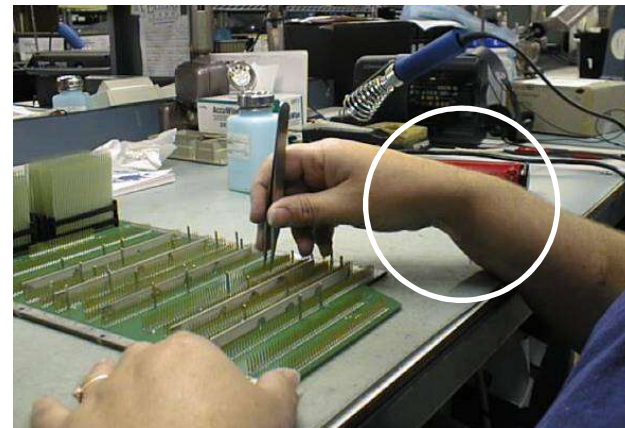
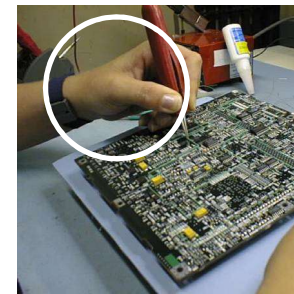
Wrists bent (flexion)



Extension

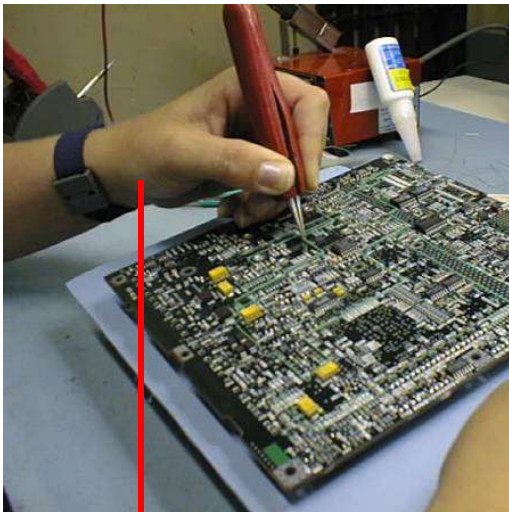


Ulnar deviation

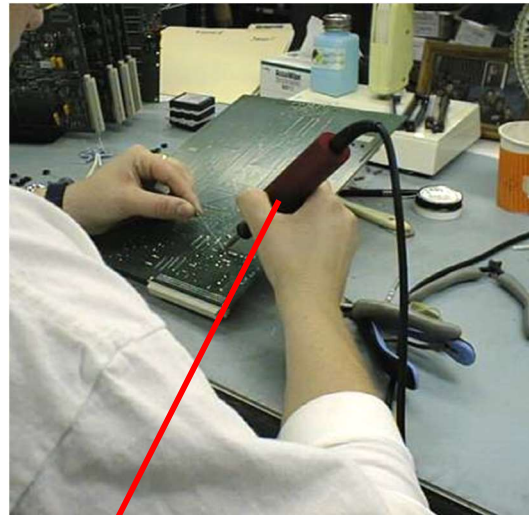


Physical Ergonomics

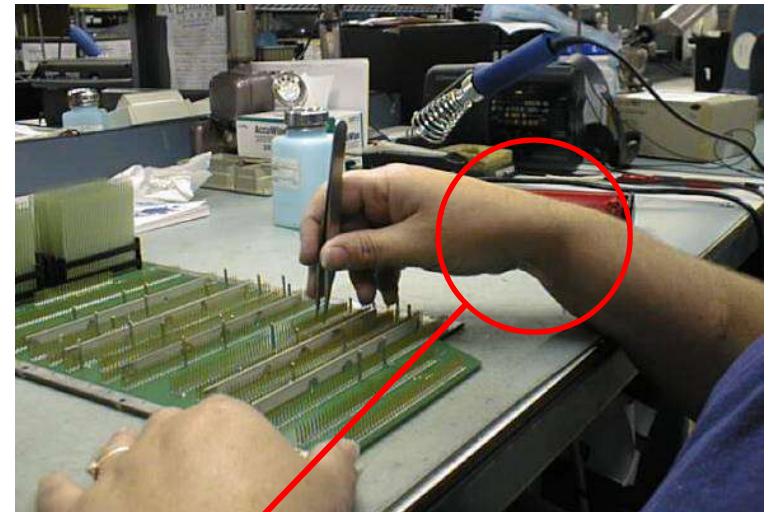
☐ Tendonitis Risk Factors of awkward / sustained postures



- Repetition



- Forceful exertion



- Mechanical stress

Physical Ergonomics

How ergonomic? Optical Mouse Pen - iPen ZDP10



ZyonSystems

- Repetition?
- Forceful exertion?
- Mechanical stress?

Physical Ergonomics

How about gaming? Repetitive Strain Injury



Fafa Yim, Logitech pro-team captain, *League of Legends*

- Repetition?
- Forceful exertion?
- Mechanical stress?

Cognitive Ergonomics

□ Cognitive

- How do users expect the technology to work?
- How is information displayed?
- How well are stereotypical expectations met?
- How complex is the interface?
- How much training is required?
- What user knowledge assumptions are met?
- How does information facilitate learning and memory?

Nuclear power plant, space travel, aviation: What are similarities cognitively?



Source: IAEA



Source: NASA



Source: Boeing

Cognitive Ergonomics

❑ Cognitive

Nuclear power plant, space travel, aviation: What are similarities cognitively?

- Complex interface: memory overload
- Life-threatening circumstances (i.e. dealing with external factors)
- Exceptionally focused attention is required
- Risk management during decision making



Source: IAEA



Source: NASA



Source: Boeing

Boeing 737 MAX 8

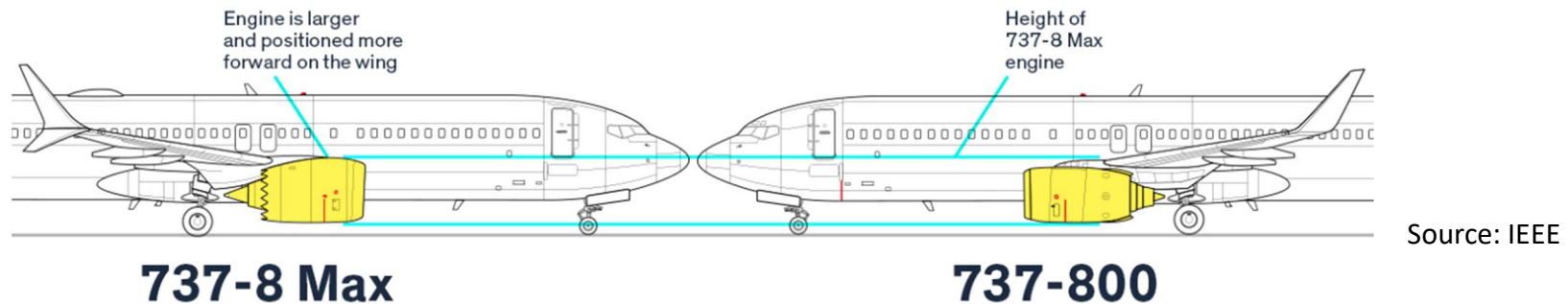


Competition > bad engineering > compensated by faulty software > improper (iPad) training personnel

Cognitive Ergonomics

Boeing 737 MAX MCAS software

- Complex interface (here, software that cannot be corrected)
- Life-threatening circumstances (here, dealing with bad engineering)
- Exceptionally focused attention is required
- Risk management during decision making



To earn a Boeing 737 type rating certificate, pilots **must** complete 21 or more days of instructor-led academics and simulator training. To be certified to fly a 737 MAX, pilots **must** either complete a 737 MAX specific type-rating course or, if a pilot is already certified to fly the 737NG, they **must** complete the NG to MAX Differences training.

Boeing emphasized that minimal training was required (saves cost for airlines)

→ MCAS software would take care of all of that?

In sum: How HCI design is indebted to Ergonomics

1. **User Orientation.** Design and application of tools, procedures, and systems must be user-oriented, rather than just “task” oriented
2. **Diversity.** Recognition of diversity in human capabilities and limitations, rather than “stereotyping” workers/users
3. **Effect on Humans.** Tools, procedures, and systems are not “inert”, but do influence human behavior and well-being
4. **Measurement.** Empirical information and evaluation is key in design process, rather than just use of “common sense”
5. **Scientific Method.** Test and retest hypothesis with real data, rather than “anecdotal” evidence or “good estimates”
6. **Systems.** Objects, procedures, environments, and people are interconnected, affect one another, and do not exist in “isolation”

Input / Output devices



Together, I/O devices form the 'user interface.' Let's look at I/O devices from an ergonomic perspective

Input devices

❑ Ergonomic keyboards

Three types:

1. Fixed split
2. Adjustable split
3. “Sculptured”

No medical evidence that they reduce Repetitive Strain Injuries... As they work for some, but not for others



Logitech ERGO K860



Kinesis Freestyle Edge RGB



Maltron Single Handed Keyboard

Awkward wrist postures minimized with 15 to 25 horizontal degree key split and 8 to 66 degree vertical incline

Input devices

☐ Pointing devices

To operate a mouse while typing, one frequently is forced to reach forward or sideways, or even both at the same time

Four types:

1. Conventional mouse / scroll wheel
2. Trackballs
3. Vertical mouse
4. 3D mouse (e.g., for 3D engineering drawings)



Logitech M570 Wireless Trackball



Logitech MX Vertical



3Dconnexion SpaceMouse Pro

Input devices

❑ Pointing devices

Four types:

1. Conventional mouse / scroll wheel
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3. Vertical mouse
4. 3D mouse (e.g., for 3D engineering drawings)

To operate a mouse while typing, one frequently is forced to reach forward or sideways, or even both at the same time

Advantage trackball over mice: Stationary, uses less space, no need to move it around, works on any type of surface, including your knee. Who outdid the trackball? Touchpad!

David Ellsworth



Logitech M570 Wireless Trackball



Logitech MX Vertical



3Dconnexion SpaceMouse Pro

Input devices

❑ Pointing devices

Pros and cons of a mouse as input (pointing) device



Pros

- works in small spaces (trackballs are better?)
- can adjust granularity of movement
- inexpensive
- user can keep eye on display
- direct relationship between hand and cursor movement on the dimensions of direction, distance, and speed
- diagonal and continuous movement, spaced control



Cons

- hand must be removed from the keyboard
- require space beside keyboard movements
- relative mode only
- mechanical mice pick up dust and other debris / optical mice might not work on certain surfaces (e.g., glass)
- require a certain amount of learned eye-hand coordination (awkward and difficult for first-time users)

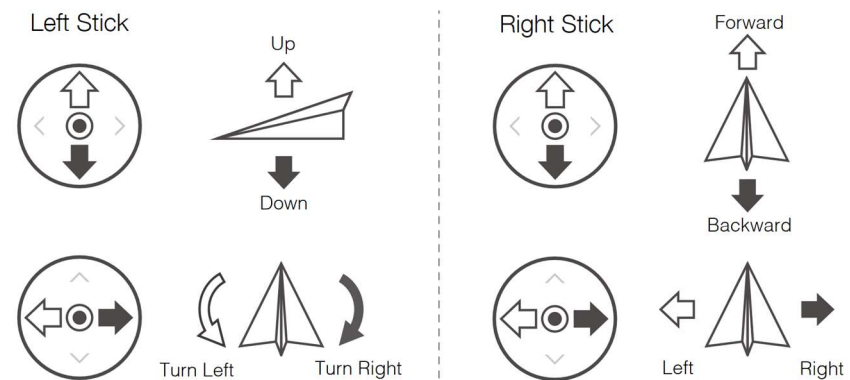


Input devices

- Degree of Freedom (DoF)
 - The limits to which physical movement or other physical processes are possible
- Joysticks (2D, like a mouse)
 - Left stick - 2 DoF
 - Right stick - 2 DoF
- Drone (3D)
 - Up / down
 - Turn left / right
 - Forward / backward
 - Move left / right



Parrot mini drone



- Left or right → X axis
- Forward (up) or back (down) → Y axis
- Rotating left (counter-clockwise) or right (clockwise) → Z axis



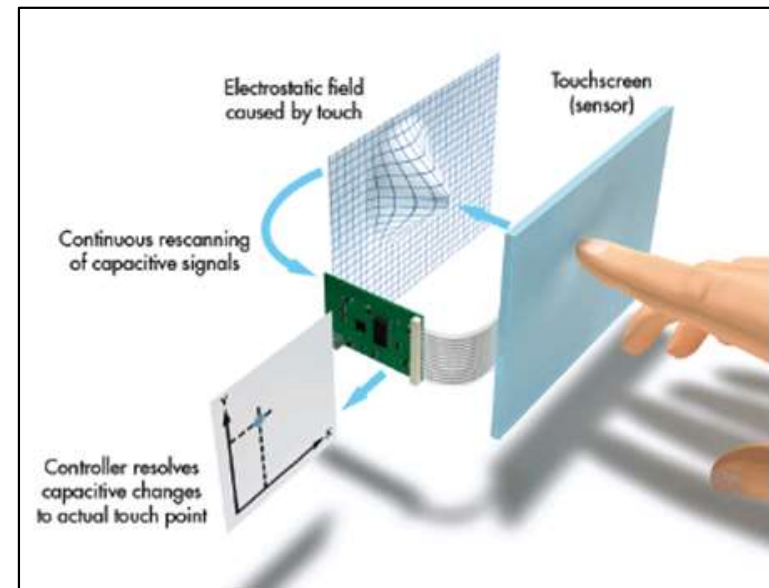
DJI Smart Controller

Cognitive ergonomics can be a problem, owing to 'translation' of representation into action + left and right hand differs

Input devices

❑ Touch screens (the pros)

- direct manipulation (i.e., direct eye-hand coordination)
- several technologies to choose from, including pressure-sensitive, resistive, infrared, **capacitive**
- faster and easier to learn than other input devices (kids can use!)
 - no command memorization needed
 - user may be led through correct command sequence
 - good for infrequent use
 - minimal training needed, high user acceptance
- continuous motion in all directions
- no extra desk space
- no moving parts



Input devices

❑ Touch screens (the cons)

- very fast, not very accurate
 - “big” fingers (Lecture 01)
 - limited resolution
 - difficult to select small targets (Fitts!)
 - very slow text and data entry
- finger/arm may obscure screen
- overlays may lead to motion parallax (Lecture 03)
- inadvertent activation
- screen can get dirty (oil from fingers)
- susceptible to temperature and humidity
- arm fatigue for conventional computer monitors
 - should be limited to low-frequency usage

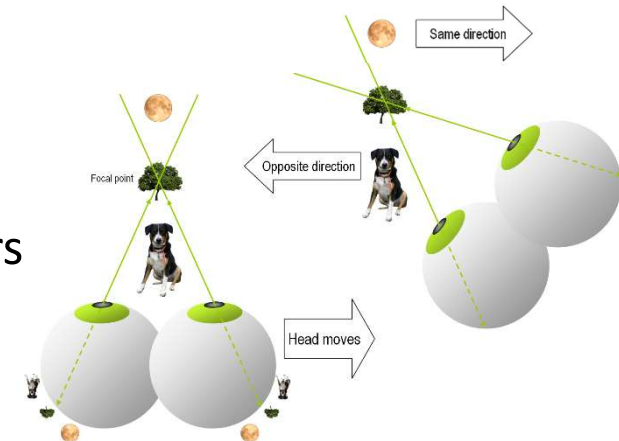
Fat finger costs Japanese broker US\$225m

(Guardian/Agencies)

Updated: 2005-12-10 06:34

$$T = a + b \log_2 \left(2 \frac{D}{W} \right)$$

Time ↓ Distance ↓
Coefficients ↑ Width ↑



Input devices

❑ Speech recognition (NLP)

Conversion of spoken language to commands or data

- Advantages over other input methods:
- More natural form of communication, so less training required
- Does not require the use of hands or other limbs, so user can carry out multiple other actions
- Opportunities for physically disabled users



Problems

- Limitations of speech recognition systems (i.e., error prone)
- Susceptible to environmental noise
- Impractical in situations where quiet is required
- Natural language capability is not yet attainable by speech recognition systems
- Speech input is not suitable for all input tasks

Input devices

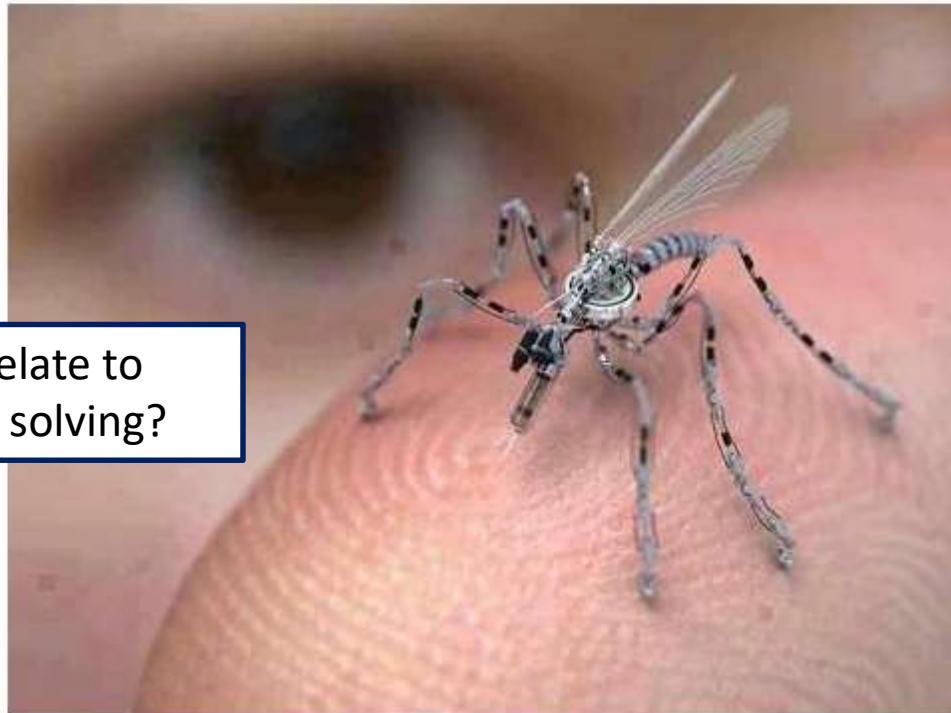
❑ Monitoring devices

- Cameras, mics, and similar devices that allow interpretation of “natural” human activities
 - observation of movements for a task
 - gestures
- Requires distinction between:
 - control activity
(signals to the computer that something relevant is going to happen)
 - intentional activity
(purposefully performed within a task)
 - spurious activity
(activity unrelated to the task at hand)



Is this a mosquito?

No. It's an insect spy drone for urban areas, already in production, funded by the US Government. It can be remotely controlled and is equipped with a camera and a microphone. It can land on you, and it may have the potential to take a DNA sample or leave RFID tracking nanotechnology on your skin. It can fly through an open window, or it can attach to your clothing until you take it in your home.



How does this relate to creative problem solving?

One of the current areas of research reportedly being undertaken in the scientific/military field is the development of micro air vehicles (MAVs), tiny flying objects intended to go places that cannot be (safely) reached by humans or other types of equipment. One of the primary military applications envisioned for MAVs is the gathering of intelligence (through the surreptitious use of cameras, microphones, or other types of sensors);

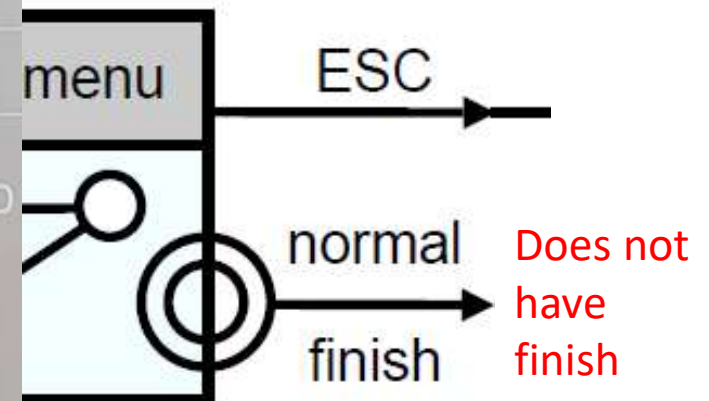
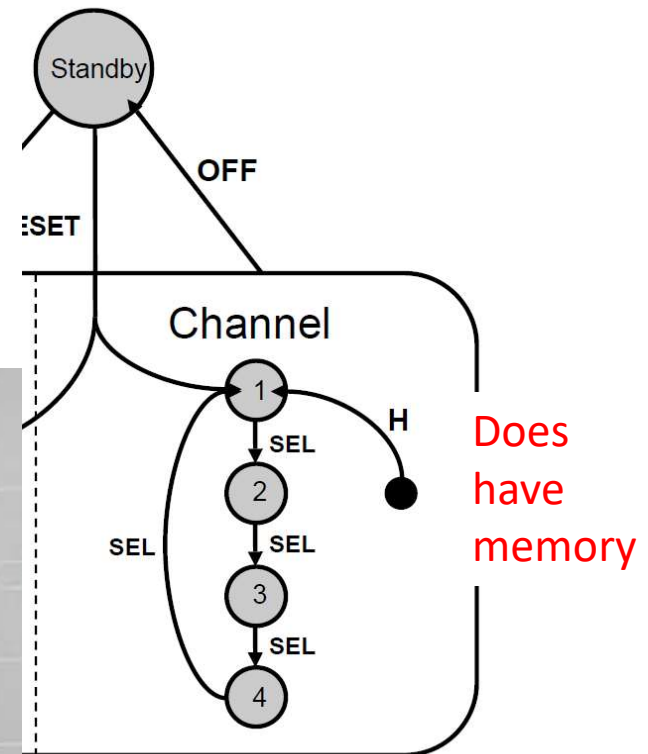
Input devices

Among all the input devices we have just discussed, is there a one suitable for people with visual impairment?



Dialog techniques

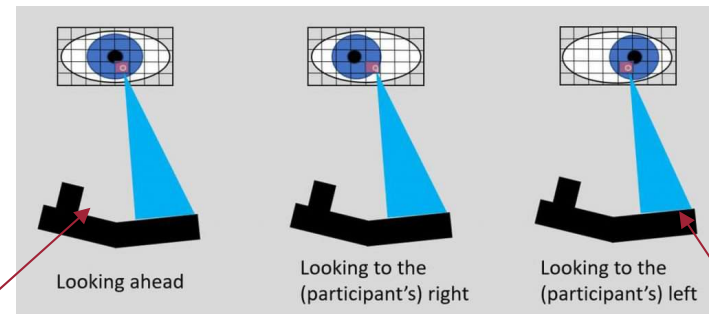
Now you know why you do dialog techniques



Input Devices

❑ Eyetracker

- Camera captures the infrared reflection of a user's cornea
- Track eyeball rotations via calculation (see figure below)
- Often used in usability and physiology experiments (for recording and analysis of user attention)
- Sometimes used as input device (e.g., selection of commands or letters by looking at them)



Camera

IR Illuminator

User must maintain a stable image on the fovea. Not easy. Not great for tracking small targets or where fine control is required (Fitts!)

Input Devices

❑ Eyetracker

Visualization of eye tracking data

- Heatmap (up)
 - Red - more fixations
 - Green - less fixations
- Scanpath (bottom)
 - Saccades - rapid, ballistic movements of the eyes that abruptly change the point of fixation
 - smooth pursuit movements - much slower tracking movements of the eyes designed to keep a moving stimulus on the fovea
 - (con)vergence movements (Lecture 04) - align the fovea of each eye with targets located at different distances from the observer
 - vestibulo-ocular movements - stabilize the eyes relative to the external world, thus compensating for head movements.



Input devices

❑ Headtracker

- Head movement to convey control information to the computer (limited range of movements)
- Keyboard is a display on the computer screen. The system detects slight movements of the user's head and moves the cursor accordingly. To operate a key, the user must locate the cursor, and then blow on a blow switch in the headset mouthpiece. This type of device can be used by severely impaired users.



Source: wired.com



Source: Intel

Input devices

❑ Headtracker

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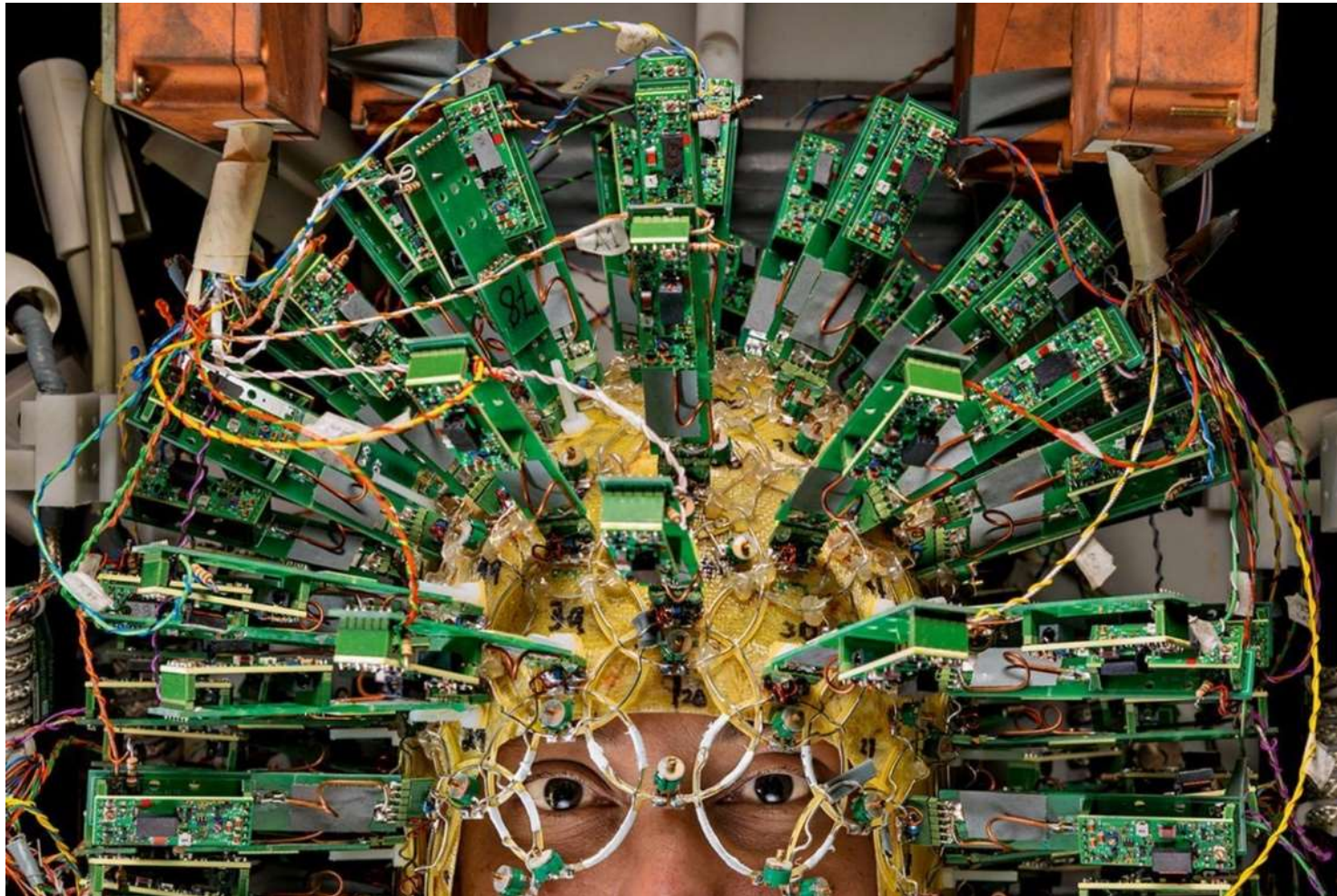
Source: wired.com



Source: Intel

Input devices

- ❑ Brain-Computer Interfaces (BCI)

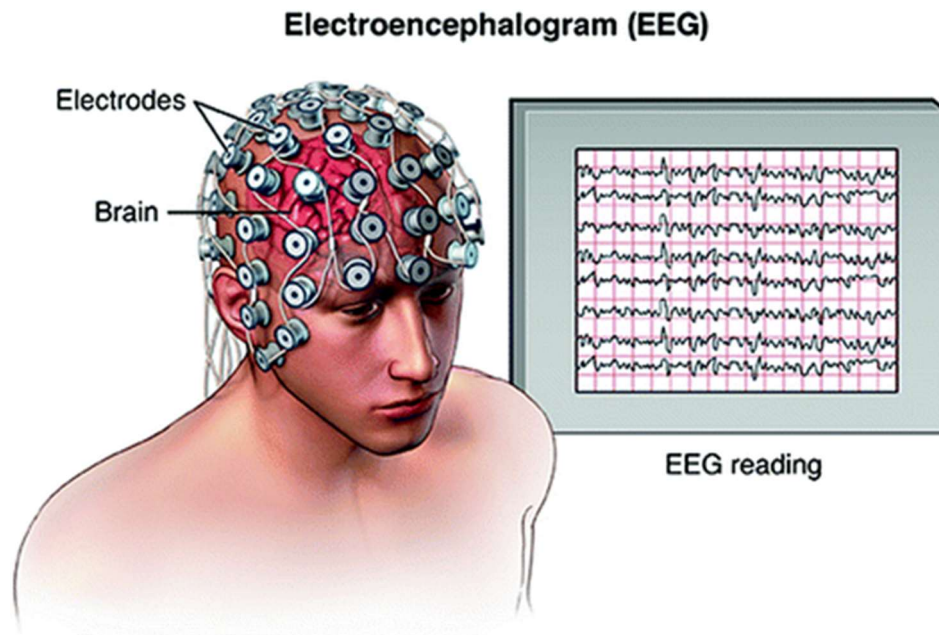


Input devices

❑ Brain-Computer Interfaces (BCI)

- Simplified versions of electroencephalograms (EEGs)

Electrodes have to be secured to the scalp/skin to detect physiological activity, and are therefore subject to general body movement (noise, clips!). Ergonomically, uncomfortable and confining. Not suited for tracking over small targets or for fine control



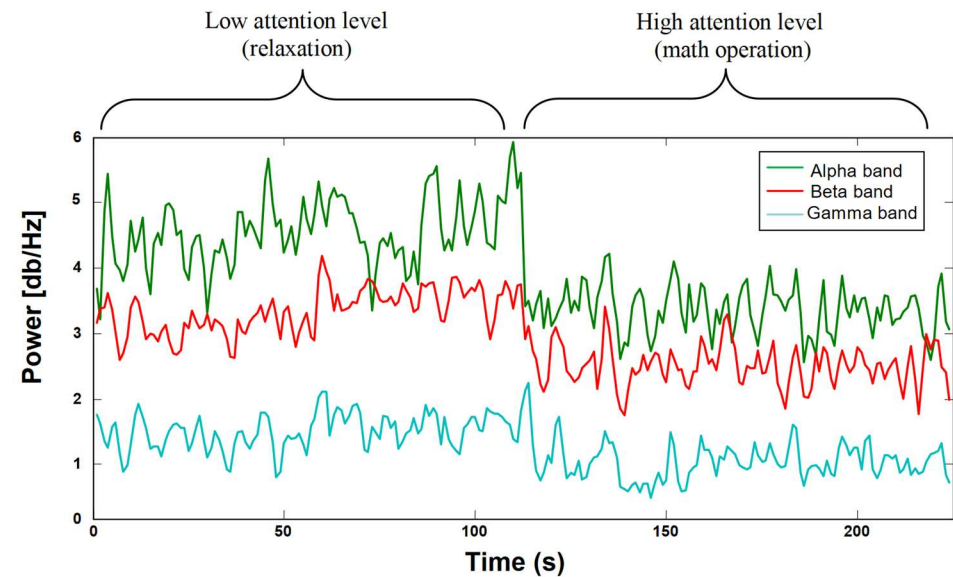
Input devices

❑ Brain-Computer Interfaces (BCI)

- Mind control through concentration of attention
- Emotion interpretation is nonsense (too context dependent)



Emotiv's wired headset EPOC has 16 contacts



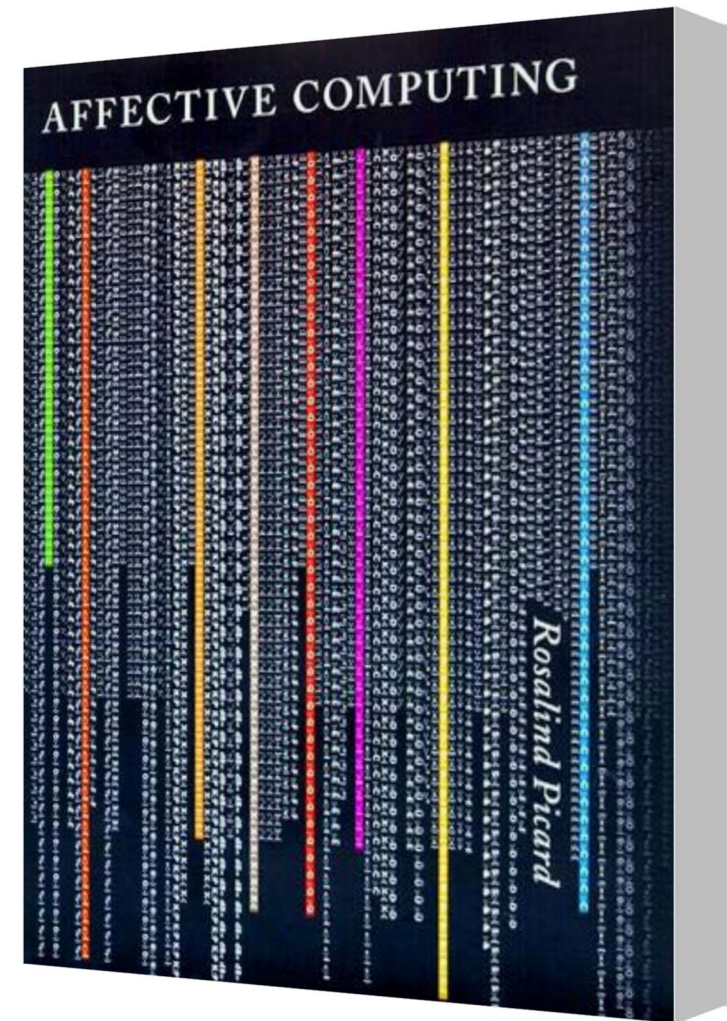
Breakout: Affective computing

Picard (1997)

Emotions affect rational thinking (e.g., decision making, perception, learning)

Computers should recognize, understand, and express emotions

Ideally, computers sample data from head pose, face, eyes (gaze), gestures, heart rate, EEG, etc. to guess the mood a user is in and respond appropriately



Breakout: Affective computing

Ring, L., Shi, L., Totzke, K., & Bickmore, T. (2015)

Conversational agent-based system

Longitudinal social support to isolated older adults

In the homes of 14 people (one week)



Agent assesses affective state via dialogue
("How are you?")

Provides empathetic feedback

Brief chats for social support

Builds sense of companionship

Story telling to encourage positive affect

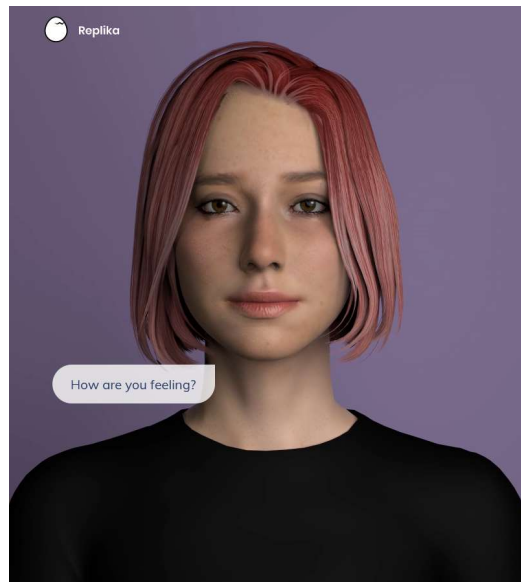
Motivational dialogue for physical activity (i.e. walking)

High levels of acceptance and satisfaction reported

Need for proactively stimulating older adults into interaction

Breakout: Affective computing

- Commercial Products
 - Replika - <https://replika.ai/>
 - Wysa - <https://www.wysa.io/>



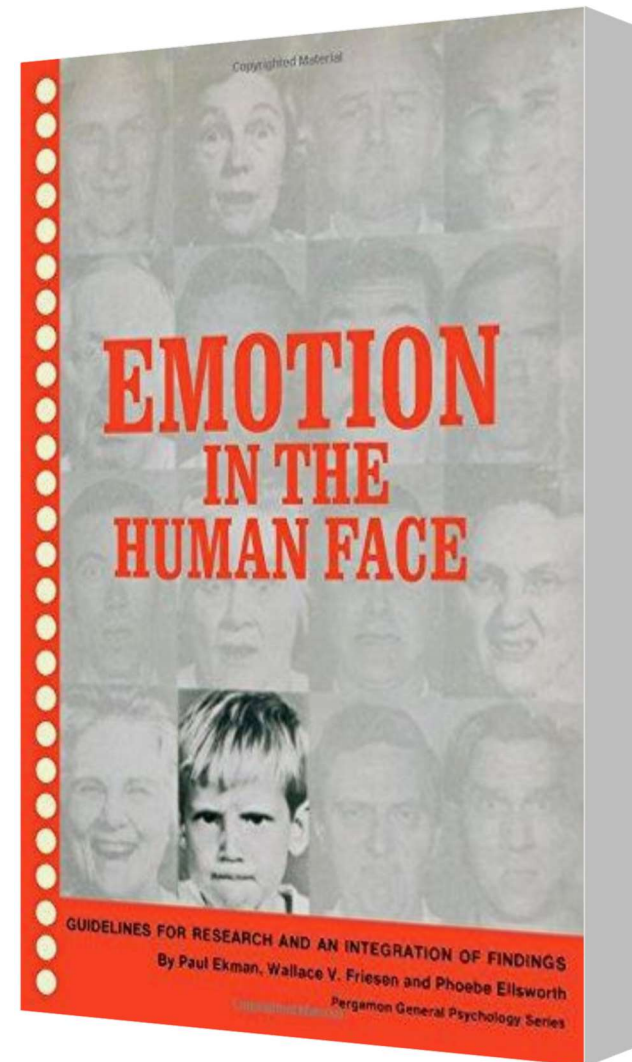
Breakout: Affective computing

Leans heavily on the work of Ekman and co-workers for emotion recognition (e.g., Ekman, Friesen, & Ellsworth, 2013)

A number of emotions may be regarded as 'universal' or 'basic' but which is still under debate (The Big 4, 5, 6, 8?)

Moreover, their expression may differ according to culture and individual

Further, interpretation of emotions is situation or context dependent!! Tough job for human, let alone for computers



Kuleshov Effect

Lev Kuleshov (1899–1970), filmmaker and theorist

Same face, same expression, different context:
What does that grin mean?



One of the first Media Psychology experiments ever

Sadness?

Disgust?

Hunger?

Desire?

Lust?

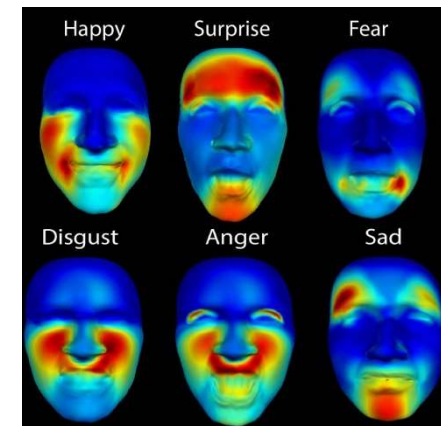
Love?

Acceptance?

Contemplation?

Cruelty?

In 90% of real-life cases, people smile out of frustration. Huge problem for any sentiment-analysis system (Hoque & Picard, 2011)



Measuring facial muscles won't help you with the interpretation

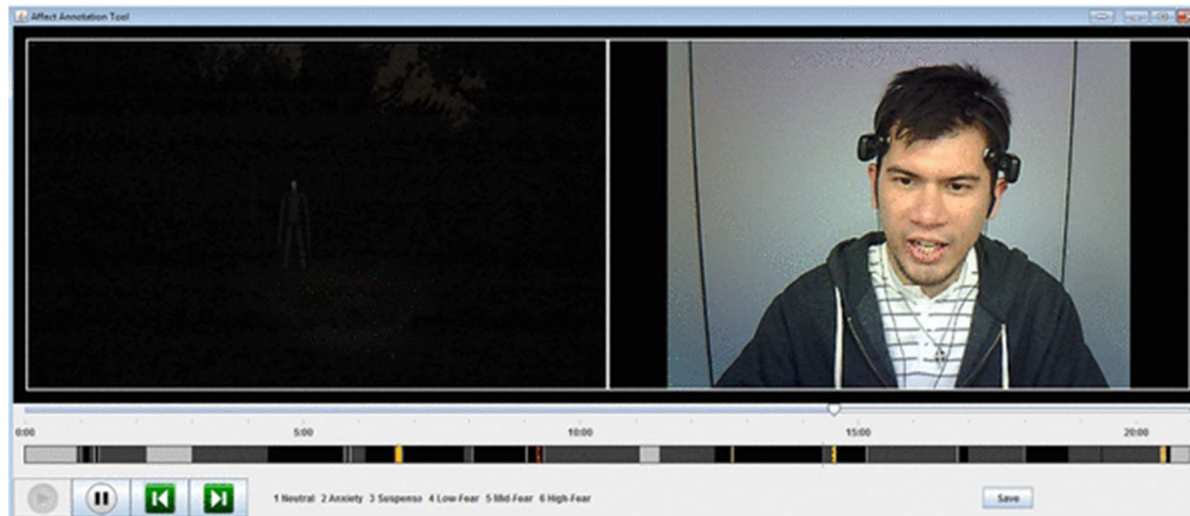
Breakout: Affective Computing

Vachiratamporn, Legaspi, Moriyama, Ichi Fukui, and Numao (2015)

Used heart rate, EEG, and keyboard activity to estimate emotions in the faces of horror-game players

Affective responses (i.e. anxiety and suspense) collected from gamers before and after scary events

Used affect annotation tool to self-report emotions after game play (to interpret physiological correlates)



Breakout: Affective Computing

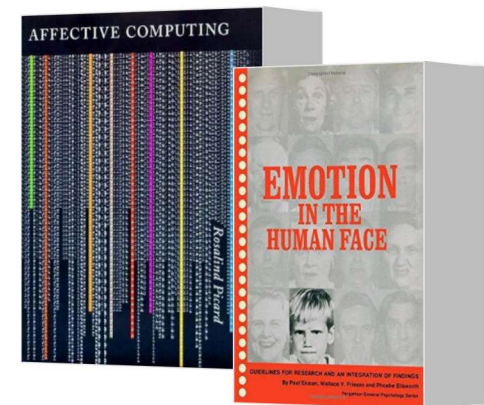
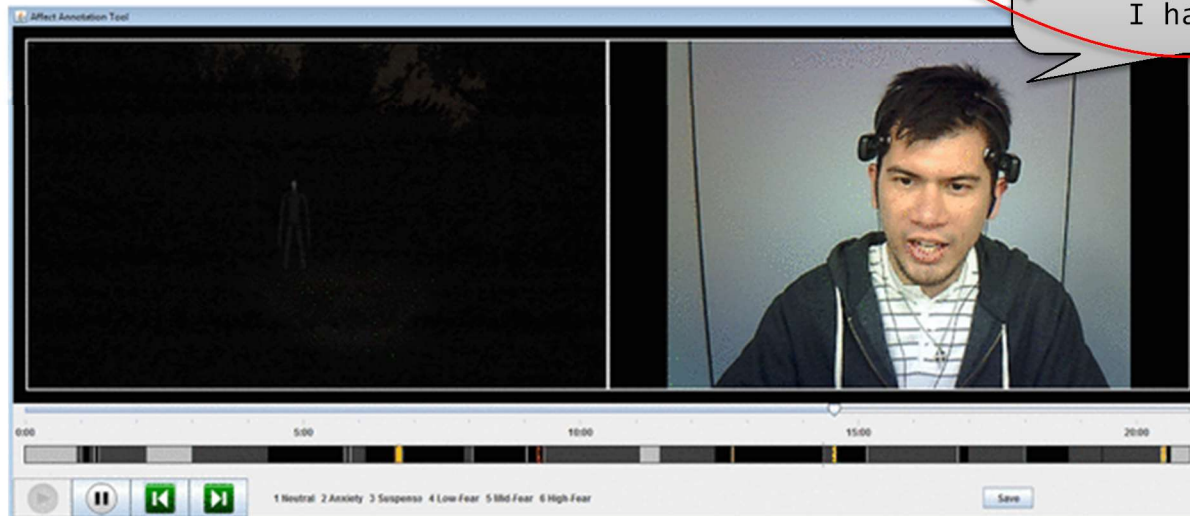


What's the problem with self-report afterwards?

Emotions

Thinking about an emotion wipes out
feeling the emotion

I think I felt
scared... or was it
frustrated? I know
I had fun!



Human-generated summary



- Ergonomics
 - Human well-being & overall system performance
 - Physical ergonomics
 - Cognitive ergonomics
- HCI largely developed from Ergonomics (Human Factors) because it is concerned with human performance and well-being in working with (computing) machinery
- I/O Devices, from a simple mouse to advanced Brain-Computer Interfaces, directly concern Ergonomics because they are the interface to the computational system. If they don't work, your system runs down
- Ergonomics is mainly known for its physical aspects, e.g., preventing mouse arms. A disaster coming from a lack of physical ergonomics was the 'big finger' incident at the Tokyo stock exchange (Lecture 01)

Human-generated summary



- Input Devices
 - Conventional input devices: keyboard, pointing devices, joysticks, touch interface
 - Recent updates: eye tracking, head tracking, BCI, VR goggles and gloves

I/O Devices may tend to all sensory channels humans have (visual, acoustic, haptic, even smell). For a device to be ergonomically functional, you have to study the human physiological, perceptual, and information processing systems (Lecture 02 and 03). Otherwise, you don't know how it works and make design errors

References

- Kuleshov Effect original video clip: https://www.youtube.com/watch?v=_gGl3LJ7vHc
- WSJ - Ergonomics Expert Explains How to Set Up Your Desk
https://www.youtube.com/watch?v=F8_ME4VwTiw
- Emotion in the Human Face (by PAUL EKMAN, WALLACE V. FRIESEN and PHOEBE ELLSWORTH)
<https://www.sciencedirect.com/book/9780080166438/emotion-in-the-human-face>

